23 Linearised Gravity and Gravitational Waves

23.1 Preliminary Remarks

While it is evidently of interest to apply general relativity to situations where the gravitational field is so strong that the Newtonian approximation fails (and we will consider such situations later on), in most ordinary situations, the gravitational field is weak, very weak, and then it is legitimate to work with a linearisation of the Einstein equations.

When we first derived the Einstein equations we checked that we were doing the right thing by deriving the Newtonian theory in the limit where

1. the gravitational field is weak
2. the gravitational field is static
3. matter is non-relativistic

The fact that the 2nd condition had to be imposed in order to recover the Newtonian theory is interesting in its own right, as it suggests that there are novel features in general relativity, even for weak fields, when the 2nd condition is not imposed. Indeed, as we will see, in this more general setting one discovers that general relativity also predicts the existence of *gravitational waves*, i.e. linearised perturbations of the gravitational field propagating like ordinary waves on the (Minkowski) background geometry. Our principal aim in this section will be to derive these linearised equations, show that they are wave equations, and study some of their more elementary consequences.

An important next step would be to study and understand how or under which circumstances gravitational waves are created and how they can be detected. These, unfortunately, are rather complicated questions in general and I will not enter into this. The things I will cover in the following are much more elementary, both technically and conceptually, than other applications of general relativity discussed elsewhere in these notes.

23.2 Linearised Einstein Equations

We express the weakness of the gravitational field by the condition that the metric be 'close' to that of Minkowski space, i.e. that

$$g_{\mu\nu} = g^{(1)}_{\mu\nu} \equiv \eta_{\mu\nu} + h_{\mu\nu} \tag{23.1}$$

with $|h_{\mu\nu}| \ll 1$. This means that we will drop terms which are quadratic or of higher power in $h_{\mu\nu}$. Here and in the following the superscript $^{(1)}$ indicates that we keep only up to linear (first order) terms in $h_{\mu\nu}$. In particular, the inverse metric is

$$g^{(1)\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu} \tag{23.2}$$

where indices are raised with $\eta^{\mu\nu}$.

As one has thus essentially chosen a background metric, the Minkowski metric, one can think of the linearised version of the Einstein equations (which are field equations for $h_{\mu\nu}$) as a Lorentz-invariant theory of a symmetric tensor field propagating in Minkowski space-time. I won't dwell on this but it is good to keep this in mind. It gives rise to the field theorist's picture of gravity as the theory of an interacting spin-2 field (which is useful for many purposes but which I do not subscribe to unconditionally because it is an inherently perturbative and background dependent picture).

It is straightforward to work out the Christoffel symbols and curvature tensors in this approximation. The terms quadratic in the Christoffel symbols do not contribute to the Riemann curvature tensor and one finds

$$\begin{aligned} \Gamma^{(1)\mu}_{\nu\lambda} &= \eta^{\mu\rho}\tfrac{1}{2}(\partial_\lambda h_{\rho\nu} + \partial_\nu h_{\rho\lambda} - \partial_\rho h_{\nu\lambda}) \\ R^{(1)}_{\mu\nu\rho\sigma} &= \tfrac{1}{2}(\partial_\rho\partial_\nu h_{\mu\sigma} + \partial_\mu\partial_\sigma h_{\rho\nu} - \partial_\rho\partial_\mu h_{\nu\sigma} - \partial_\nu\partial_\sigma h_{\rho\mu}) \end{aligned} \tag{23.3}$$

(this result for the Riemann tensor can also be inferred directly from the expression (8.13) of the Riemann tensor at the origin of an inertial coordinate system). Hence

- the linearised Ricci tensor is

$$R^{(1)}_{\mu\nu} = \tfrac{1}{2}(\partial_\sigma\partial_\nu h^\sigma_{\ \mu} + \partial_\sigma\partial_\mu h^\sigma_{\ \nu} - \partial_\mu\partial_\nu h - \Box h_{\mu\nu}) \ , \tag{23.4}$$

  where

$$h \equiv h^\mu_{\ \mu} = \eta^{\mu\nu}h_{\mu\nu} = -h_{00} + \delta^{ik}h_{ik} \tag{23.5}$$

  is the trace of $h_{\mu\nu}$ and

$$\Box = \partial^\mu\partial_\mu = -(\partial_0)^2 + \Delta \tag{23.6}$$

  is the Minkowski wave operator;

- the linearised Ricci scalar is

$$R^{(1)} = \eta^{\mu\nu}R^{(1)}_{\mu\nu} = \partial_\mu\partial_\nu h^{\mu\nu} - \Box h \ ; \tag{23.7}$$

- the linearised Einstein tensor is

$$\begin{aligned} G^{(1)}_{\mu\nu} &= R^{(1)}_{\mu\nu} - \tfrac{1}{2}\eta_{\mu\nu}R^{(1)} \\ &= \tfrac{1}{2}(\partial_\sigma\partial_\nu h^\sigma_{\ \mu} + \partial_\sigma\partial_\mu h^\sigma_{\ \nu} - \partial_\mu\partial_\nu h - \Box h_{\mu\nu} - \eta_{\mu\nu}\partial_\rho\partial_\sigma h^{\rho\sigma} + \eta_{\mu\nu}\Box h) \ ; \end{aligned} \tag{23.8}$$

- and the linearised vacuum Einstein equations

$$R^{(1)}_{\mu\nu} = 0 \quad \Leftrightarrow \quad G^{(1)}_{\mu\nu} = 0 \tag{23.9}$$

  are thus

$$R^{(1)}_{\mu\nu} = 0 \quad \Leftrightarrow \quad \partial_\sigma\partial_\nu h^\sigma_{\ \mu} + \partial_\sigma\partial_\mu h^\sigma_{\ \nu} - \partial_\mu\partial_\nu h - \Box h_{\mu\nu} = 0 \ , \tag{23.10}$$

  or, equivalently (in the form $G^{(1)}_{\mu\nu} = 0$)

$$\partial_\sigma\partial_\nu h^\sigma_{\ \mu} + \partial_\sigma\partial_\mu h^\sigma_{\ \nu} - \partial_\mu\partial_\nu h - \Box h_{\mu\nu} - \eta_{\mu\nu}\partial_\rho\partial_\sigma h^{\rho\sigma} + \eta_{\mu\nu}\Box h = 0 \ . \tag{23.11}$$

The latter can be derived from the quadratic Minkowski space (Poincaré invariant) field theory action (the *Fierz-Pauli action* (1939))

$$\begin{aligned} S[h_{\mu\nu}] &= \int d^4x\, \mathcal{L}(h_{\mu\nu}, \partial_\sigma h_{\mu\nu}) \\ \mathcal{L}(h_{\mu\nu}, \partial_\sigma h_{\mu\nu}) &= -\tfrac{1}{4} h_{\mu\nu,\sigma} h^{\mu\nu,\sigma} + \tfrac{1}{2} h_{\mu\nu,\sigma} h^{\sigma\mu,\nu} + \tfrac{1}{4} h^{,\sigma} h_{,\sigma} - \tfrac{1}{2} h_{,\sigma} h^{\mu\sigma}{}_{,\mu} \end{aligned} \tag{23.12}$$

for a free massless spin-2 field described by the Lorentz tensor $h_{\mu\nu}$. Indeed, variation of the action with respect to the $h_{\mu\nu}$ (and the usual integration by parts) leads to

$$h_{\mu\nu} \to h_{\mu\nu} + \delta h_{\mu\nu} \quad \Rightarrow \quad \delta S[h_{\mu\nu}] = -\int d^4x\, G^{(1)}_{\mu\nu} \delta h^{\mu\nu} \ . \tag{23.13}$$

Remarks:

1. Signs 1: note that from (23.2) one has

   $$g^{(1)\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu} \quad \Rightarrow \quad \delta g^{(1)\mu\nu} = -\delta h^{\mu\nu} \ , \tag{23.14}$$

   so that the sign in (23.13) is the standard one if expressed in terms of the variation of $g^{\mu\nu}$,

   $$\delta S[h_{\mu\nu}] = +\int d^4x\, G^{(1)}_{\mu\nu} \delta g^{(1)\mu\nu} \ . \tag{23.15}$$

2. Signs 2: the sign and overall normalisation of the Lagrangian are also such that one obtains the canonically normalised kinetic term

   $$\mathcal{L} = +\tfrac{1}{4} (\partial_t h_{ik})^2 + \dots \tag{23.16}$$

   for a 2-index field.

3. Note that the Lagrangian is *not* the linearised Einstein-Hilbert Lagrangian, i.e. the linearised Ricci scalar (23.7) (as the latter is, by construction, linear in $h_{\mu\nu}$). Rather,

   $$\mathcal{L} = \tfrac{1}{2} h^{\mu\nu} G^{(1)}_{\mu\nu} \ + \ \text{total derivative} \ , \tag{23.17}$$

   where the total derivative contributions serve to turn the terms of the form $h\partial^2 h$ arising from $h^{\mu\nu} G^{(1)}_{\mu\nu}$ into the standard $(\partial h)^2$-form, so that $\mathcal{L}$ is of a standard (quadratic) form in the derivatives of $h_{\mu\nu}$, as it should be.

4. Note that we can indeed take the integration measure to be the flat Minkowski measure d^4x, as $\mathcal{L}$ is already quadratic in $h_{\mu\nu}$ and its derivatives, so that any contribution of $h_{\mu\nu}$ to the measure would give a subleading contribution.

5. This theory, just like its spin-1 Maxwell counterpart, has a local gauge invariance which we will return to and make use of below.

6. In the presence of matter the linearised Einstein equations are

$$G^{(1)}_{\mu\nu} = 8\pi G_N T^{(0)}_{\mu\nu} \quad . \tag{23.18}$$

Note that only the zero'th order term in the h-expansion appears on the right hand side of this equation. This is due to the fact that $T_{\mu\nu}$ must itself already be small in order for the linearised approximation to be valid, i.e. $T^{(0)}_{\mu\nu}$ should be of order $h_{\mu\nu}$. Therefore, any terms in $T_{\mu\nu}$ depending on $h_{\mu\nu}$ would already be of order $(h_{\mu\nu})^2$ and can be dropped.

7. The conservation law for the energy-momentum tensor is therefore just

$$\partial_\mu T^{(0)\mu\nu} = 0 \quad , \tag{23.19}$$

and this is indeed compatible with the linearised Bianchi identity

$$\partial_\mu G^{(1)\mu\nu} = 0 \quad , \tag{23.20}$$

which can easily be verified, and which reflects the invariance of the theory under the linearised coordinate transformations to be discussed below.

23.3 Newtonian Limit Revisited

In section 19.3 we verified that the Newtonian (weak field, static, non-relativistic matter) limit of the Einstein equations reduces to the Newtonian (Poisson) equation $\Delta\phi = 4\pi G_N \rho$. This is a special case of the above general weak-field limit equations (23.18), with $G^{(1)}_{\mu\nu}$ given in (23.8), and it will be instructive to redo the calculation of section 19.3 from this more general perspective.

We thus assume an energy-momentum tensor whose only non-negligible component is the energy density $T_{00} = \rho$, with ρ static and $G_N\rho \ll 1$. Then we can assume that the deviations of the space-time geometry from the Minkowski metric are small and time-independent,

$$g_{\mu\nu} = g^{(1)}_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \quad , \quad \partial_0 h_{\mu\nu} = 0 \quad . \tag{23.21}$$

Then the (00)-component of the Einstein tensor reduces (after some cancellations) to

$$G^{(1)}_{00} = -\tfrac{1}{2}\Delta(\delta^{ik}h_{ik}) + \tfrac{1}{2}\partial_i\partial_k h^{ik} \quad . \tag{23.22}$$

In particular, h_{00} and its derivatives have dropped out of this expression, which appears to be at odds with the desired $G_{00} = -\Delta g_{00}$ for Newtonian fields (section 19.3). However, we have not yet used at all the condition that $T_{ik} = 0 \to G_{ik} = 0$. In particular, for static perturbations we find from (23.8) that the trace of the spatial components of the Einstein tensor is

$$\delta^{ik}G^{(1)}_{ik} = \tfrac{1}{2}\Delta(\delta^{ik}h_{ik}) - \tfrac{1}{2}\partial_i\partial_k h^{ik} - \Delta h_{00} \tag{23.23}$$

so that

$$G^{(1)}_{ik} = 0 \quad \Rightarrow \quad G^{(1)}_{00} = -\Delta h_{00} \ . \tag{23.24}$$

This is precisely the relation required (and verified) in the analysis of section 19, in order to have the correct Newtonian limit.

23.4 ADM and Komar Energies of an Isolated System

As we are in the realm of a Poincaré-invariant classical field theory, we can define the total energy of the system as the integral of the energy-density $T_{00} = \rho$ over a spatial constant time $x^0 = \text{const.}$ slice Σ,

$$E = \int_\Sigma d^3x \; T_{00} \tag{23.25}$$

(another way to see that we can take the integration measure to be the flat Euclidean measure d^3x is to recall that T_{00} is small by assumption). If the linearised Einstein equations are satisfied, we can express the total energy in terms of the Einstein tensor,

$$E = \frac{1}{8\pi G_N} \int_\Sigma d^3x \; G^{(1)}_{00} \ . \tag{23.26}$$

Now we have two distinct expressions for $G^{(1)}_{00}$ in terms of the derivatives of the metric and, interestingly, both of them are spatial total derivatives, namely

$$G^{(1)}_{00} = -\tfrac{1}{2}\Delta(\delta^{ik}h_{ik}) + \tfrac{1}{2}\partial_i\partial_k h^{ik} = \partial_i(-\tfrac{1}{2}\partial^i h^k_{\ k} + \tfrac{1}{2}\partial_k h^{ik}) \tag{23.27}$$

and

$$G^{(1)}_{00} = -\Delta h_{00} = -\partial_i(\partial^i h_{00}) \ . \tag{23.28}$$

This allows us to write the total energy E of the system as a boundary integral over the boundary

$$\partial\Sigma = S^2_\infty \tag{23.29}$$

of the spatial slice "at infinity" in two different ways, namely as

$$E = \frac{1}{16\pi G_N} \oint_{S^2_\infty} dS_i \; (\partial_k h^{ik} - \partial^i h^k_{\ k}) \tag{23.30}$$

or as

$$E = -\frac{1}{8\pi G_N} \oint_{S^2_\infty} dS_i \; \partial^i h_{00} \ . \tag{23.31}$$

Note that these final expressions only depend on the asymptotics of the gravitational field at spatial infinity (the assumption that the gravitational field is weak there translating into the statement that the metric is asymptotically flat). It is thus tempting to adopt E as the definition of the total energy of an isolated (i.e. asymptotically flat) system in general, even when the field in the interior is not weak:

1. Thus in general the *ADM Energy* of such a system is defined by

$$E_{ADM} = \frac{1}{16\pi G_N} \oint_{S^2_\infty} dS_i \, (\partial_k h_{ik} - \partial_i h_{kk}) \ , \tag{23.32}$$

this expression relying on a Cartesian coordinate system at infinity (which is why we can permit ourselves to be careless with the positioning of the indices - repeated indices are to be summed, as usual). Somewhat more covariantly one can write this as

$$E_{ADM} = \frac{1}{16\pi G_N} \oint_{S^2_\infty} dS_i \, (\bar{\nabla}^k h_{ik} - \bar{\nabla}_i h) \ , \tag{23.33}$$

where $\bar{\nabla}$ is the covariant derivative with respect to a background metric $\bar{g}_{ik}$,

$$h_{ik} = g_{ik} - \bar{g}_{ik} \ , \tag{23.34}$$

and

$$h = \bar{g}^{ik} h_{ik} \tag{23.35}$$

is the trace of h_{ik} with respect to the background metric. If the background metric is the Minkowski metric, *and* one works in Cartesian coordinates, then (23.33) reduces to (23.32).

The justification for the identification of this quantity with the energy is substantially strengthened by the canonical (Hamiltonian) ADM analysis of general relativity discussed in section 21, in particular in section 21.12. Indeed, in section 21.12 we deduced another candidate expression for the energy from the boundary term in the ADM Hamiltonian, namely (21.191)

$$E_{ADM} = -\frac{1}{8\pi G_N} \oint_{S_\infty} d^2x \, \sqrt{s}(k_S - k^0_S) \tag{23.36}$$

At first sight, beyond the fact that both are expressed as integrals over a 2-sphere at infinity and that both depend in some way on a choice of reference (background) metric, these two expresssions appear to have little in common. However, it is possible to show that (23.33) is in fact equal to (23.36) provided that the metrics on S_∞ induced by the metrics $\bar{g}_{ik}$ (appearing in (23.33)) and g^0_{ik} (implicitly appearing in (23.36)) are the same. The proof of this statement I am aware of relies on a convenient choice of coordinate system and is therefore not particularly insightful *per se* and will not be given here.[64]

2. The second expression is actually a special case of the Komar charges briefly mentioned in section 13.7, here applied to the (asymptotic) timelike Killing vector ∂_t (the generator of time-translations, and thus naturally associated with the

[64] For this proof, see e.g. S. Hawking, G. Horowitz, *The Gravitational Hamiltonian, Action, Entropy, and Surface Terms*, `arXiv:gr-qc/9501014`, or Exercise 4.5.7 of E. Poisson, *A Relativist's Toolkit* (modelled on the cited article).

energy or Hamiltonian). Indeed, recall that for any Killing vector K^μ we had the conserved current $J^\mu = R^\mu_{\ \nu} K^\nu$ (13.49), which was itself a divergence of an anti-symmetric tensor, $J^\mu = \nabla_\nu A^{\mu\nu}$, with $A^{\mu\nu} = -A^{\nu\mu} = \nabla^\mu K^\nu$. Thus the corresponding conserved charge $Q_K(V)$ contained in a volume V can be written as a surface integral of $\nabla^\mu K^\nu$ over the boundary ∂V of the volume V,

$$Q_K(V) \sim \oint_{\partial V} dS_{\mu\nu} \nabla^\mu K^\nu \ . \tag{23.37}$$

Applying this to $V = \Sigma$, $\partial V = S^2_\infty$, and $K = \partial_t$, we find (asymptotically)

$$\begin{aligned} \nabla^i K^0 &= \nabla_i K^0 = \Gamma^0_{i\alpha} K^\alpha = \Gamma^0_{i0} = -\Gamma_{0i0} = -\tfrac{1}{2}\partial_i h_{00} \\ \nabla^0 K^i &= -\nabla_0 K^i = -\Gamma^i_{0\alpha} K^\alpha = -\Gamma^i_{00} = -\Gamma_{i00} = +\tfrac{1}{2}\partial_i h_{00} \ . \end{aligned} \tag{23.38}$$

Thus the second expression (23.31) can be written as

$$\begin{aligned} E &= -\frac{1}{8\pi G_N} \oint_{S^2_\infty} dS_i \ \partial_i h_{00} = -\frac{1}{4\pi G_N} \oint_{S^2_\infty} dS_{0i} \ \nabla^0 K^i \\ &= -\frac{1}{8\pi G_N} \oint_{S^2_\infty} dS_{\mu\nu} \ \nabla^\mu K^\nu \ . \end{aligned} \tag{23.39}$$

This fixes the normalisation of the Komar charge $Q_K(V)$ (23.37) in this case,

$$E_{\text{Komar}}(V) \equiv Q_{K=\partial_t}(V) = -\frac{1}{8\pi G_N} \oint_{\partial V} dS_{\mu\nu} \ \nabla^\mu K^\nu \tag{23.40}$$

and we can identify (23.31) with the total Komar energy ($V = \Sigma$) associated to the Killing vector $K = \partial_t$,

$$E_{\text{Komar}}(\Sigma) = -\frac{1}{8\pi G_N} \oint_{S^2_\infty} dS_i \ \partial_i h_{00} = -\frac{1}{8\pi G_N} \oint_{S^2_\infty} dS_{\mu\nu} \ \nabla^\mu K^\nu \ . \tag{23.41}$$

As our primary litmus test, in section 24.8 we will apply these expressions to the Schwarzschild metric, the unique spherically symmetric asymptotically flat solution of the vacuum Einstein equations. In particular, it therefore describes the gravitational field in the exterior of a star, and it depends on a single parameter m, related to the mass of the star by $m = G_N M/c^2$, and it is then of interest to see what the ADM and Komar energies have to say about this. Suffice it to say here that reassuringly one indeed finds $E_{ADM} = E_{\text{Komar}} = M$.

In the same way one can also introduce various notions of momentum or angular momentum of an isolated system (the latter for instance being non-zero for the Kerr metric (30.3) describing a rotating star or black hole), but we will not pursue this here.

23.5 Wave Equations and Gauge Conditions in Maxwell Theory

We will now abandon the assumption of static non-relativistic fields and return to the general weak-field linearised Einstein equations (23.18). In order to understand how to proceed from there, in this section we will briefly recall in a condensed way the analogous steps in the case of Maxwell theory.

1. The Maxwell equations (in a Minkowski background) are already linear (no need to linearise) and read

$$\partial^\alpha F_{\alpha\beta} = -J_\beta \ . \tag{23.42}$$

In terms of the potentials A_α these equations are

$$\Box A_\beta - \partial_\beta(\partial_\alpha A^\alpha) = -J_\beta \ . \tag{23.43}$$

These are to be thought of as the analogue of the linearised Einstein equations (23.18), with the linearised Einstein tensor given in (23.8). These are wave equations (with a source) for the components A_α, but due to the second term on the left-hand side the individual components of the potential are not yet completely decoupled in this equation.

2. Maxwell theory has the gauge invariance

$$A_\alpha \to A_\alpha + \partial_\alpha V \tag{23.44}$$

(in particular, the field strenghts $F_{\alpha\beta}$ are invariant) which allows one to choose for instance the Loren(t)z gauge condition[65]

$$\partial_\alpha A^\alpha = 0 \ . \tag{23.45}$$

Indeed, if V is chosen to satisfy the differential equation

$$\Box V = -\partial_\alpha A^\alpha \ , \tag{23.46}$$

then

$$\partial^\alpha(A_\alpha + \partial_\alpha V) = 0 \ . \tag{23.47}$$

With this choice of gauge the Maxwell equations become decoupled wave equations,

$$\Box A_\beta = -J_\beta \ , \tag{23.48}$$

and can now be straightforwardly solved in terms of Green functions etc. One particular solution to the inhomogeneous equation is the retarded potential

$$A_\alpha(t,\vec{x}) = (4\pi)^{-1} \int d^3x' \, \frac{J_\alpha(t-|\vec{x}-\vec{x}'|,\vec{x}')}{|\vec{x}-\vec{x}'|} \ . \tag{23.49}$$

Note that, as a consequence of $\partial_\alpha J^\alpha = 0$, this solution is automatically in the Lorenz gauge. As usual, the general solution is then a sum of this particular solution of the inhomogeneous equation and the general solution of the homogeneous equation.

[65] Credit for this should really go the Danish physicist Ludvig V. Lorenz (who used this condition in its Lorent(!)z non-invariant form already in 1867, just 3 years after the publicaton of Maxwell's treatise) rather than the more famous Dutch physicist Hendrik A. Lorentz. See J. Jackson's *Electrodynamics* (note at the end of chapter 6).

3. The homogeneous equation, i.e. the vacuum Maxwell equations, can now be solved in terms of plane waves,

$$A_\alpha = \epsilon_\alpha \mathrm{e}^{ik_\beta x^\beta} \quad \text{with} \quad k_\beta k^\beta = 0 \tag{23.50}$$

(or wave packets constructed from them), and the Lorenz gauge constrains the *polarisation vector* ϵ_α by

$$k^\alpha \epsilon_\alpha = 0 \ . \tag{23.51}$$

For a wave travelling in the x^3-direction, this implies for the polarisation vector

$$k^\alpha = (\omega, 0, 0, k^3 = \omega) \quad \Rightarrow \quad \epsilon_3 = -\epsilon_0 \ . \tag{23.52}$$

4. One still has the residual gauge invariance

$$A'_\alpha = A_\alpha + \partial_\alpha \xi \quad , \quad \Box \xi = 0 \tag{23.53}$$

leaving the Lorenz gauge condition invariant. Choosing a particular solution of this homegenous wave equation, say

$$\xi = \xi_0 \mathrm{e}^{ik_\beta x^\beta} \tag{23.54}$$

now completely fixes this residual gauge invariance. Under this residual gauge transformation the polarisation vector transforms as

$$\epsilon'_\alpha = \epsilon_\alpha + i\xi_0 k_\alpha \ , \tag{23.55}$$

in particular

$$\epsilon'_0 = \epsilon_0 - i\xi_0 \omega \ . \tag{23.56}$$

Thus choosing $\xi_0 = \epsilon_0 / i\omega$, one has $\epsilon'_0 = 0$, implying $\epsilon'_3 = 0$, and one is left with the polarisation vector

$$\epsilon^\alpha = (0, \epsilon^1, \epsilon^2, 0) \tag{23.57}$$

displaying the two physical transverse polarisations of an electromagnetic wave.

23.6 Linearised Gravity: Gauge Invariance and Coordinate Choices

We now proceed analogously in the case of linearised gravity. First of all we need to understand the gauge invariance (or the counterpart of gauge invariance) in the present case.

The original non-linear Einstein equations have a local invariance consisting of general coordinate transformations. What remains of general coordinate invariance in the linearised approximation are, naturally, linearised general coordinate transformations. Indeed, $h_{\mu\nu}$ and

$$h'_{\mu\nu} = h_{\mu\nu} + L_V \eta_{\mu\nu} \tag{23.58}$$

represent the same physical perturbation because $\eta_{\mu\nu} + L_V \eta_{\mu\nu}$ is just an infinitesimal coordinate transform of the Minkowski metric $\eta_{\mu\nu}$. Therefore linearised gravity has the gauge freedom

$$h_{\mu\nu} \to h_{\mu\nu} + \partial_\mu V_\nu + \partial_\nu V_\mu \ . \tag{23.59}$$

For example, the linearised Riemann tensor $R^{(1)}_{\mu\nu\rho\sigma}$ is, rather obviously, invariant under this transformation. As a consequence, in particular also the Einstein tensor, and hence the vacuum linearised Einstein equations, have this local gauge invariance. This can be understood without any calculation by noting that under the variation

$$\delta_V h_{\mu\nu} = L_V \eta_{\mu\nu} \tag{23.60}$$

the linearised Riemann tensor transforms into the Riemann tensor for the Minkowski metric,

$$\delta_V R^{(1)}_{\mu\nu\rho\sigma} = L_V R_{\mu\nu\rho\sigma}(\eta_{\alpha\beta}) = 0 \ . \tag{23.61}$$

However, in contrast to the Lagrangian of Maxwell theory, say, the Lagrangian (23.12) for linearised gravity is *not* strictly gauge invariant, but only invariant up to a total derivative.

Given this gauge invariance of the linearised theory, for explicit calculations it is useful to make a particular gauge choice and, as in Maxwell theory, a good choice of gauge can simplify things considerably (and a bad choice of gauge can have the opposite effect).

It turns out that, in general, a very useful gauge condition is

$$g^{\mu\nu}\Gamma^\rho_{\mu\nu} = 0 \ . \tag{23.62}$$

It is called the *harmonic gauge condition*, or Fock, or de Donder gauge condition (even though harmonic coordinates were used extensively by Einstein himself until just before the discovery of the final, truly generally covariant, formulation of general relativity). The name harmonic derives from the fact that in this gauge the coordinate functions x^μ are harmonic:

$$\Box x^\mu \equiv g^{\nu\rho}\nabla_\rho \partial_\nu x^\mu = -g^{\nu\rho}\Gamma^\mu_{\nu\rho} \ , \tag{23.63}$$

and thus

$$\Box x^\mu = 0 \ \Leftrightarrow \ g^{\nu\rho}\Gamma^\mu_{\nu\rho} = 0 \ . \tag{23.64}$$

It is the analogue of the Lorenz gauge $\partial_\mu A^\mu = 0$ in Maxwell theory. Moreover, in flat space Cartesian coordinates are obviously harmonic, and in general harmonic coordinates are (like geodesic coordinates) a nice and useful curved space counterpart of Cartesian coordinates.

In the linearised theory, this gauge condition becomes

$$\partial_\mu h^\mu_\lambda - \tfrac{1}{2}\partial_\lambda h = 0 \ . \tag{23.65}$$

The gauge parameter V_μ which will achieve this is the solution to the equation

$$\Box V_\lambda = -(\partial_\mu h^\mu_\lambda - \tfrac{1}{2}\partial_\lambda h) \ . \tag{23.66}$$

Indeed, with this choice one has

$$\partial_\mu(h^\mu_\lambda + \partial^\mu V_\lambda + \partial_\lambda V^\mu) - \tfrac{1}{2}\partial_\lambda(h + 2\partial^\mu V_\mu) = 0 \ . \tag{23.67}$$

Note for later that, as in Maxwell theory, this gauge choice does not necessarily fix the gauge completely. Any transformation $x^\mu \to x^\mu + \xi^\mu$ with $\Box\xi^\mu = 0$ will leave the harmonic gauge condition invariant.

23.7 Wave Equation

Now let us use this gauge condition in the linearised Einstein equations. In this gauge they simplify somewhat to

$$\Box h_{\mu\nu} - \tfrac{1}{2}\eta_{\mu\nu}\Box h = -16\pi G_N T^{(0)}_{\mu\nu} \ . \tag{23.68}$$

In particular, the vacuum equations (or the equations in a source-free region of space-time) are just

$$T^{(0)}_{\mu\nu} = 0 \ \ \Rightarrow \ \ \Box h_{\mu\nu} = 0 \ , \tag{23.69}$$

which is the standarad relativistic wave equation. Together, the equations

$$\begin{aligned} \Box h_{\mu\nu} &= 0 \\ \partial_\mu h^\mu_\lambda - \tfrac{1}{2}\partial_\lambda h &= 0 \end{aligned} \tag{23.70}$$

determine the evolution of a disturbance in a gravitational field in vacuum in the harmonic gauge.

It is often convenient to consider the linear combination

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \tfrac{1}{2}\eta_{\mu\nu}h \ . \tag{23.71}$$

This is also commonly known as the *trace reversed perturbation*, because in 4 space-time dimensions (but only there) one has

$$\bar{h}^\mu_\mu = -h^\mu_\mu \ . \tag{23.72}$$

Note, as an aside, that with this notation and terminology the Einstein tensor (again in 4 space-time dimensions only) is the trace reversed Ricci tensor,

$$\bar{R}_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R = G_{\mu\nu} \ . \tag{23.73}$$

In terms of $\bar{h}_{\mu\nu}$, the linearised Einstein equations and the harmonic gauge condition (in any dimension) are just

$$\begin{aligned} \Box\bar{h}_{\mu\nu} &= -16\pi G_N T^{(0)}_{\mu\nu} \\ \partial_\mu \bar{h}^\mu_{\ \nu} &= 0 \ . \end{aligned} \tag{23.74}$$

This way of writing the linearised Einstein equations sharpens the analogy with the Maxwell equations in the Lorenz gauge:

- the Lorenz gauge $\partial_\mu A^\mu = 0$ decouples the Maxwell equations for A_μ which in this gauge read $\Box A_\mu = -j_\mu$
- the gauge condition $\partial_\mu \bar{h}^\mu_{\ \nu} = 0$ decouples the linearised Einstein equations for the variables $\bar{h}_{\mu\nu}$ which in this gauge read $\Box\bar{h}_{\mu\nu} = -16\pi G_N T^{(0)}_{\mu\nu}$.

One solution is, of course, the retarded solution

$$\bar{h}_{\mu\nu}(t,\vec{x}) = 4G_N \int d^3x' \ \frac{T^{(0)}_{\mu\nu}(t-|\vec{x}-\vec{x}'|,\vec{x}')}{|\vec{x}-\vec{x}'|} \ . \tag{23.75}$$

Note that, as a consequence of $\partial_\mu T^{(0)\mu\nu} = 0$, this solution is automatically in the harmonic gauge. As usual, the general solution is then a sum of this particular solution of the inhomogeneous equation and the general solution of the homogeneous equation.

23.8 Polarisation Tensor and the Metric of a Gravitational Wave

The homogeneous equation is the linearised vacuum Einstein equation in the harmonic gauge,

$$\Box\bar{h}_{\mu\nu} = 0 \ . \tag{23.76}$$

This is clearly solved by

$$\bar{h}_{\mu\nu} = \epsilon_{\mu\nu} \mathrm{e}^{ik_\alpha x^\alpha} \ , \tag{23.77}$$

where $\epsilon_{\mu\nu}$ is a constant, symmetric *polarisation tensor* and k^α is a constant *wave vector*, provided that k^α is null, $k^\alpha k_\alpha = 0$. In order to obtain real metrics from this one should of course use real linear combinations of such solutions in the end.

Thus plane waves are solutions to the linearised equations of motion and the Einstein equations predict the existence of gravitational waves travelling along null geodesics (at the speed of light). The timelike component of the wave vector is often referred to as the *frequency* ω of the wave, and we can write $k^\mu = (\omega, k^i)$. Plane waves are of course not the most general solutions to the wave equations but any solution can be written as a superposition of plane wave solutions (wave packets).

So far, we have ten parameters $\epsilon_{\mu\nu}$ and four parameters k^μ to specify the wave, but many of these are spurious, i.e. can be eliminated by using the freedom to perform linearised coordinate transformations and Lorentz rotations.

First of all, the harmonic gauge condition implies that

$$\partial_\mu \bar{h}^\mu_{\ \nu} = 0 \quad \Rightarrow \quad k^\mu \epsilon_{\mu\nu} = 0 \ . \tag{23.78}$$

Now we can make use of the residual gauge freedom $x^\mu \to x^\mu + \xi^\mu$ with $\Box \xi^\mu = 0$ to impose further conditions on the polarisation tensor. Since this is a wave equation for ξ^μ, once we have specified a solution for ξ^μ we will have fixed the gauge completely. Alternatively, note that under $h_{\mu\nu} \to h_{\mu\nu} + \partial_\mu V_\nu + \partial_\nu V_\mu$ one has

$$\begin{aligned} & \bar{h}_{\mu\nu} \to \bar{h}_{\mu\nu} + \partial_\mu V_\nu + \partial_\nu V_\mu - \eta_{\mu\nu} \partial_\lambda V^\lambda \\ \Rightarrow \quad & \partial^\mu \bar{h}_{\mu\nu} \to \partial^\mu \bar{h}_{\mu\nu} + \Box V_\nu \end{aligned} \tag{23.79}$$

so that the gauge condition is invariant precisely under linearised coordinate transformations with $\Box V_\mu = 0$. Taking the solution of this equation to be of the form

$$V_\mu = v_\mu \mathrm{e}^{ik_\alpha x^\alpha} \ , \tag{23.80}$$

the polarisation tensor transforms according to

$$\epsilon_{\alpha\beta} \to \epsilon_{\alpha\beta} + i(k_\alpha v_\beta + k_\beta v_\alpha) - i\eta_{\alpha\beta} k^\gamma v_\gamma \ . \tag{23.81}$$

One can now choose the v_μ in such a way (see the example below) that the new polarisation tensor satisfies $k^\mu \epsilon_{\mu\nu} = 0$ (as before) as well as

$$\epsilon_{\mu 0} = \epsilon^\mu_{\ \mu} = 0 \ . \tag{23.82}$$

All in all, we appear to have nine conditions on the polarisation tensor $\epsilon_{\mu\nu}$ but as both (23.78) and the first of (23.82) imply $k^\mu \epsilon_{\mu 0} = 0$, only eight of these are independent. Therefore, there are two independent polarisations for a gravitational wave.

Together with $k^\mu \epsilon_{\mu\nu} = 0$, these gauge conditions thus impose the conditions

$$k^\mu \bar{h}_{\mu\nu} = 0 \quad , \quad \bar{h}_{\mu 0} = 0 \quad , \quad \bar{h}^\mu_{\ \mu} = 0 \ . \tag{23.83}$$

This is known as the *transverse traceless gauge*, and a field satisfying this gauge is frequently denoted by $\bar{h}^{TT}_{\mu\nu}$.

For example, let us consider a wave travelling in the x^3-direction,

$$k^\mu = (\omega, 0, 0, k^3) = (\omega, 0, 0, \omega) \ . \tag{23.84}$$

Then

- the condition $k^\mu \epsilon_{\mu\nu} = 0$ becomes $\epsilon_{3\nu} = -\epsilon_{0\nu}$

- the condition $\epsilon_{00} = 0$ determines $v_0 - v_3$, and also implies $\epsilon_{30} = 0$
- the other linear combination $v_0 + v_3 \sim k^\alpha v_\alpha$ can be used to achieve $\epsilon^\mu_{\ \mu} = 0$
- the conditions $\epsilon_{10} = \epsilon_{20} = 0$ determine v_1 and v_2

Therefore, the only independent components are ϵ_{ab} with $a, b = 1, 2$. As ϵ_{ab} is symmetric and traceless, this wave is completely characterised by $\epsilon_{11} = -\epsilon_{22}$, $\epsilon_{12} = \epsilon_{21}$ and the frequency ω.

Now we should not forget that, when talking about the polarisation tensor of a gravitational wave, we are actually talking about the space-time metric itself. Namely, since for a traceless perturbation we have

$$\epsilon^\mu_{\ \mu} = 0 \quad \Rightarrow \quad \bar{h}_{\alpha\beta} = h_{\alpha\beta} \ , \tag{23.85}$$

we have deduced that the metric describing a gravitational wave travelling in the x^3-direction can always be put into the form

$$ds^2 = -dt^2 + (\delta_{ab} + h_{ab}) dx^a dx^b + (dx^3)^2 \ , \tag{23.86}$$

with $h_{ab} = h_{ab}(t \mp x^3)$ symmetric and traceless. This neatly encodes and describes the distortion of the space-time geometry in the directions transverse to the gravitational wave.

Remarks:

1. Introducing null coordinates

$$u = t - x^3 \quad , \quad v = t + x^3 \ , \tag{23.87}$$

the gravitational wave metric can be written as

$$ds^2 = -du\, dv + (\delta_{ab} + h_{ab}(u)) dx^a dx^b \ . \tag{23.88}$$

Explicitly, with $x^a = (x, y)$, the linearised perturbation can be written as

$$h_{ab}(u) dx^a dx^b = h_+(u)(dx^2 - dy^2) + 2h_\times(u) dx dy \ , \tag{23.89}$$

with two arbitrary functions $h_{11}(u) \equiv h_+(u)$ and $h_{12}(u) \equiv h_\times(u)$ (see the next section 23.9 for an explanation of this notation).

2. This analysis was rather evidently independent of the dimension. In D dimensions the polarisation states of a graviton are described by a symmetric, transverse, and traceless tensor h_{ab} where $a, b = 1, \ldots, D-2$. Thus the number of physical polarisation states of a graviton in D dimensions are

$$\#[h_{ab}] = \frac{(D-2)(D-1)}{2} - 1 = \frac{D(D-3)}{2} \ . \tag{23.90}$$

Note that this gives zero for $D = 3$, in agreement with the fact, noted before, that in 3 space-time dimensions there is no gravitational vacuum dynamics since vanishing of the Ricci tensor is equivalent to vanishing of the Riemann tensor.

3. By abandoning the assumption that $h_{ab}(u)$ be small, one can look for (and easily find) solutions of the full non-linear Einstein equations of the form

$$ds^2 = -du\, dv + g_{ab}(u)dx^a dx^b \quad . \tag{23.91}$$

These are known as exact gravitational plane waves, and are discussed in some detail in section 43.

23.9 Physical Effects of Gravitational Waves

To determine the physical effect of a gravitational wave racing by, we cannot just look at the gravitational field (23.86) at a point (by the equivalence principle), i.e. we cannot detect the presence of such a wave (ultra-)locally. However, we can consider its influence on the relative motion of nearby particles. In other words, we look at the geodesic deviation equation (8.38).

Consider a family of nearby particles described by the *velocity field* $u^\mu(x)$ and separation (deviation) vector $S^\mu(x)$. Then the change of the deviation vector along the flow lines of the velocity field is determined by

$$(D_\tau)^2 S^\mu = R^\mu{}_{\nu\rho\sigma} u^\nu u^\rho S^\sigma \quad . \tag{23.92}$$

We consider the situation where the test particles are initially, in the absence of the gravitational wave, at rest, $u^\mu = (1, 0, 0, 0)$. Then the gravitational wave will lead, to lowest order in the perturbation $h_{\mu\nu}$, to a 4-velocity

$$u^\mu = (1, 0, 0, 0) + \mathcal{O}(h) \quad . \tag{23.93}$$

However, because the Riemann tensor is already of order h, to lowest order the right hand side of the geodesic deviation equation reduces to

$$R^{(1)}_{\mu 00\sigma} = \tfrac{1}{2}\partial_0\partial_0 h_{\mu\sigma} \tag{23.94}$$

(because $h_{0\mu} = 0$). On the other hand, to lowest order the left hand side is just the ordinary time derivative. Thus the geodesic deviation equation becomes (an overdot denoting a t-derivative)

$$\ddot{S}^\mu = \tfrac{1}{2}\ddot{h}^\mu_\sigma\, S^\sigma \quad . \tag{23.95}$$

In particular, we see immediately that the gravitational wave is *transversally polarised*, i.e. the component S^3 of S^μ in the longitudinal direction of the wave is unaffected and

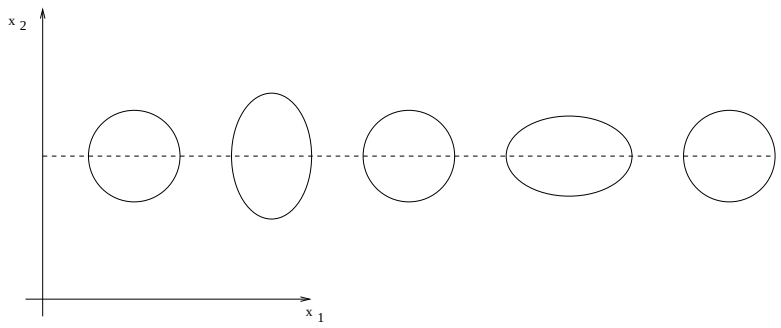


Figure 9: Effect of a gravitational wave with polarisation ϵ_{11} moving in the x^3-direction, on a ring of test particles in the $x^1 - x^2$-plane.

the particles are only disturbed in directions perpendicular to the wave. The movement of the particles in the 1-2 plane is then governed by

$$\ddot{S}^a = \tfrac{1}{2}\ddot{h}^a_{\ b}\, S^b \equiv -(\Omega^2)^a_{\ b}\, S^b \ , \tag{23.96}$$

which is the equation of a 2-dimensional time-dependent harmonic oscillator, giving rise to characteristic oscillating movements of the test particles in the 1-2 plane. With

$$h^a_{\ b} = \mathrm{e}^{\,-i\omega(t\,-\,x^3)}\epsilon^a_{\ b} \ , \tag{23.97}$$

as above, one has

$$(\Omega^2)^a_{\ b} = \tfrac{1}{2}\omega^2 h^a_{\ b} \ , \tag{23.98}$$

and we can consider separately the two cases (1) $\epsilon_{12} = 0$ and (2) $\epsilon_{11} = -\epsilon_{22} = 0$.

1. For $\epsilon_{12} = 0$ one has

$$\begin{aligned} \ddot{S}^1(t) &= -\tfrac{1}{2}\epsilon_{11}\omega^2 \mathrm{e}^{\,-i\omega t} S^1(t)\, \mathrm{e}^{\,i\omega x^3} \\ \ddot{S}^2(t) &= +\tfrac{1}{2}\epsilon_{11}\omega^2 \mathrm{e}^{\,-i\omega t} S^2(t)\, \mathrm{e}^{\,i\omega x^3} \end{aligned} \tag{23.99}$$

   Recalling that ϵ_{ab} is small, the solution to lowest order is simply

$$\begin{aligned} S^1(t) &= (1 + \tfrac{1}{2}\epsilon_{11}\mathrm{e}^{\,-i\omega(t\,-\,x^3)})S^1(0) \\ S^2(t) &= (1 - \tfrac{1}{2}\epsilon_{11}\mathrm{e}^{\,-i\omega(t\,-\,x^3)})S^2(0) \end{aligned} \tag{23.100}$$

   Given the interpretation of S^μ as a separation vector, this means that particles originally separated in the x^1-direction will oscillate back and forth in the x^1-direction and likewise for x^2. A nice (and classical) way to visualise this (see Figure 9) is to start off with a ring of particles in the $1-2$ plane. As the wave passes by the particles will start bouncing in such a way that the ring bounces in the shape of a cross $+$. For this reason, ϵ_{11} is also frequently written as ϵ_+.

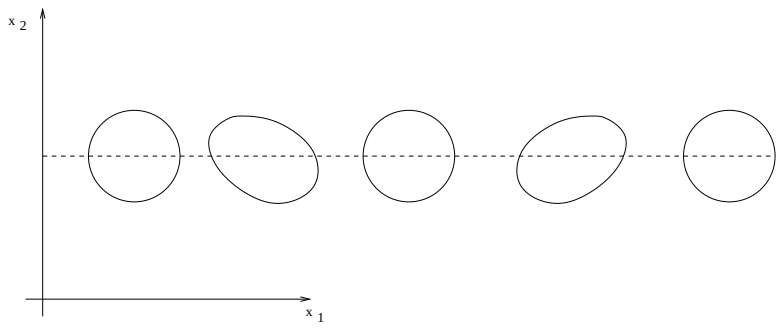


Figure 10: Effect of a gravitational wave with polarisation ϵ_{12} moving in the x^3-direction, on a ring of test particles in the $x^1 - x^2$-plane.

2. If, on the other hand, $\epsilon_{11} = 0$ but $\epsilon_{12} = \epsilon_{21} \neq 0$, then the lowest order solution is

$$\begin{aligned} S^1(t) &= S^1(0) + \tfrac{1}{2}\epsilon_{12} \mathrm{e}^{-i\omega(t-x^3)} S^2(0) \\ S^2(t) &= S^2(0) + \tfrac{1}{2}\epsilon_{12} \mathrm{e}^{-i\omega(t-x^3)} S^1(0) \quad . \end{aligned} \tag{23.101}$$

   This time the deplacement in the x^1-direction is governed by the original deplacement in the x^2-direction and vice-versa, and the ring of particles will bounce in the shape of a $\times$ ($\epsilon_{12} = \epsilon_\times$) - see Figure 10.

3. Of course, one can also construct circularly polarised waves by using

$$\epsilon_{R,L} = \frac{1}{\sqrt{2}}(\epsilon_{11} \pm i\epsilon_{12}) \quad . \tag{23.102}$$

These solutions display the characteristic behaviour of *quadrupole radiation*, and this is something that we might have anticipated on general grounds. First of all, we know from Birkhoff's theorem (see section 24.6) that a spherically symmetric vacuum solution of the Einstein field equations is necessarily static. Thus there can be no radial oscillations, and thus no monopole (s-wave) radiation. Moreover, dipole radiation is due to oscillations of the center of charge. While this is certainly possible for electric charges, an oscillation of the center of mass would violate momentum conservation and is therefore ruled out. Thus the lowest possible mode of gravitational radiation is quadrupole radiation, just as we have found.

23.10 Brief Comments on Production and Energy of Gravitational Waves

Now that we have found the solutions to the vacuum equations, we should include sources and study the production of gravitational waves, characterise the type of radiation that is emitted, estimate the radiated energy etc. However, this is quite a delicate

and both technically and conceptually quite challenging subject, and I will just develop this to the extent that the quadrupole property of the radiation becomes plausible.[66]

In order to study the production of gravitational waves, we need to include sources, i.e. we need to go back to the retarded solution (23.75)

$$\bar{h}_{\mu\nu}(t,\vec{x}) = 4G_N \int d^3y \, \frac{T^{(0)}_{\mu\nu}(t-|\vec{x}-\vec{y}|,\vec{y})}{|\vec{x}-\vec{y}|} \quad . \tag{23.103}$$

At large distances, and if the source does not oscillate too rapidly (the wavelength should be much larger than the size of the source), one can approximate this by

$$\bar{h}_{\mu\nu}(t,\vec{x}) \approx \frac{4G_N}{r} \int d^3y \, T^{ret}_{\mu\nu}(t,\vec{y}) \quad , \tag{23.104}$$

where $r = |\vec{x}|$ and the retarded source is

$$T^{ret}_{\mu\nu}(t,\vec{y}) = T^{(0)}_{\mu\nu}(t-r,\vec{y}) \quad . \tag{23.105}$$

This is the gravitational analogue of the dipole approximation to the multipole expansion in electrodynamics (and, as we will see, here this turns out to be a quadrupole approximation).

Next, since $T^{(0)}_{\mu\nu}$ is conserved, also

$$\bar{h}_{\mu 0} \sim \int d^3y \, T^{(ret)}_{\mu 0} \tag{23.106}$$

is conserved, i.e. time-independent. Therefore, in this approximation the leading $(1/r)$-part of $\bar{h}_{\mu 0}$ will not lead to gravitational waves. We can thus concentrate on the spatial components

$$\bar{h}_{ik}(t,\vec{x}) \approx \frac{4G_N}{r} \int d^3y \, T^{ret}_{ik}(t,\vec{y}) \quad , \tag{23.107}$$

Using the Laue theorem (7.53),

$$\int d^3x \, T^{ik} = \tfrac{1}{2}(\partial_0)^2 \int d^3x \, T_{00} x^i x^k \equiv \tfrac{1}{2}\ddot{Q}^{ik} \quad , \tag{23.108}$$

we thus have

$$\bar{h}_{ik}(t,\vec{x}) \approx \frac{2G_N}{r} \ddot{Q}^{ret}_{ik} \quad , \tag{23.109}$$

where (7.54)

$$Q^{ret}_{ik}(t) = \int d^3x \, \rho^{ret} x_i x_k \tag{23.110}$$

is the quadrupole moment tensor of the retarded energy density $T^{ret}_{00} = \rho^{ret}$. Thus, if the source has a time-dependence

$$\rho(t) \sim \mathrm{e}^{-i\Omega t} \quad , \tag{23.111}$$

[66] See e.g. S. Weinberg, *Gravitation and Cosmology*, B. Schutz, *A first course in general relativity*, or S. Carroll, *Spacetime and Geometry*, and of course C. Misner, K. Thorne, J. Wheeler, *Gravitation* for detailed discussionss.

say (of course, one should in the end take real superpositions of such modes), then

$$\bar{h}_{ik}(t,r) \approx -2G_N\Omega^2 Q^{ret}_{ik}\ \frac{\mathrm{e}^{-i\Omega(t-r)}}{r} \tag{23.112}$$

clearly describes an outgoing spherical wave.

As noted before, the retarded solution is automatically in the harmonic gauge, but it is not yet in the transverse traceless gauge. Transforming the above solution to the transverse traceless gauge, one finds that the (transverse, traceless) components

$$\bar{h}^{TT}_{ab} = h^{TT}_{ab} \tag{23.113}$$

are naturally expressed not in terms of the quadrupole moments Q_{ik} but in terms of the so-called "reduced" (traceless) quadrupole moments

$$\begin{aligned} \mathcal{Q}^{ret}_{ik} &= \int d^3x\ \rho^{ret}(x_i x_k - \tfrac{1}{3}\delta_{ik} r^2) \\ &= Q^{ret}_{ik} - \tfrac{1}{3}\delta_{ik}(Q^{ret})^j{}_j \ . \end{aligned} \tag{23.114}$$

These formulae can now in principle be applied to various specific situations of interest by specifying the source term appropriately.

Finally, one quantity of particular interest is of course the energy radiated away by the source. However, as discussed in section 22.6, the notion of "gravitational energy" or "energy of the gravitational field" is not in general well defined and raises numerous conceptual and technical issues. One might perhaps have hoped that these issues can be completely side-stepped in the linearised theory we are dealing with here, which is after all much more like a standard field theory in Minkowski space. And indeed, several strategies are available, and they all lead to expressions for the energy-density which are of the standard form "quadratic in the derivatives of the fields". For example one can

- proceed by analogy with quadrupole radiation in Maxwell theory,
- use the (Belinfante-improved) Noether energy-momentum tensor of the quadratic Fierz-Pauli action (23.12),
- expand the Einstein equations not only to linear but to quadratic order in the fluctuations $h_{\mu\nu}$ and interpret the quadratic terms as the gravitational contribution to the energy-momentum tensor,
- ...

E.g. in the transverse traceless gauge the Fierz-Pauli action reduces to a "standard" quadratic action

$$\mathcal{L} = -\tfrac{1}{2}(\eta^{\alpha\beta}\partial_\alpha h_{11}\partial_\beta h_{11} + \eta^{\alpha\beta}\partial_\alpha h_{12}\partial_\beta h_{12}) \ . \tag{23.115}$$

For $h_{ab} = h_{ab}(t - x^3)$, say, this gives rise to an energy density and energy flux

$$\Theta_{00} = -\Theta_{03} = (\dot{h}_{11})^2 + (\dot{h}_{12})^2 \sim r^{-2}(\dddot{\mathcal{Q}}^{ret})^2 \quad . \tag{23.116}$$

On the basis of such considerations one might expect or anticipate the total radiated energy to be proportional to something like

$$dE/dt \sim -(\dddot{Q}^{ret})^2 \quad . \tag{23.117}$$

However, dealing with quadratic terms in a linearised theory is somewhat dodgy and not strictly speaking internally consistent. As a consequence, equally plausible strategies may not necessarily lead to equivalent results. Nevertheless, there appears to be some consensus that a formula like this is indeed correct, and more specifically that (with certain approximations and averaging) one has the remarkable formula

$$\frac{dE}{dt} = -\frac{G_N}{5}(\dddot{\mathcal{Q}}^{ret})_{ik}(\dddot{\mathcal{Q}}^{ret})^{ik} \quad . \tag{23.118}$$

While this formula (with its 3rd derivative squared) may look unfamiliar, it is precisely analogous to the corresponding formula for the radiated power of an electric quadrupole in Maxwell theory (also proportional to $\dddot{Q}^2$). The main difference between gravitational and electro-magentic radiation lies in the fact that in the Maxwell case the leading (lowest multipole) contribution arises from dipole radiation, while in the gravitational case the leading contribution is quadrupole radiation.

23.11 Even Briefer Comments on Detection of Gravitational Waves

I will conclude this section with some very general comments on the detection of gravitational waves.

In principle, this ought to be straightforward. In practice, however, because of the extreme weakness of gravitational fields, this is about as far from straightforward as one can possibly imagine. For example, on the basis of the calculations done in section 23.9, one might like to simply try to track the separation of two freely suspended masses (and this is indeed part of the principle of the interferometers I will briefly return to below).

Alternatively, the particles need not be free but could be connected by a solid piece of material. Then gravitational tidal forces will stress the material. If the resonant frequency of this "antenna" equals the frequency of the gravitational wave, this should lead to a detectable oscillation. This is the principle of the so-called *Weber detectors* or *Weber bars* (1966-...). While fine in principle, in practice gravitational waves are extremely weak. To the best of my knowledge, such detectors have not produced conclusive results so far, and other detection techniques are favoured in modern generations of detectors.

More sensitive modern experiments are not fine-tuned to a particular resonant frequency but can in principle detect a continuous range of frequencies. These use detectors based on huge laser Michelson interferometers (arms several kilometers long), e.g. LIGO (Laser Interferometer Gravitational Wave Observatory) and VIRGO. These or their upgrades, or the space-based LISA (Laser Interferometer Space Antenna), are widely expected to have reached sufficient sensitivity to finally directly detect gravitational waves in the next couple of years.[67] **See the 2016 Update at the end of this section!**

However, in spite of the absence of direct evidence for gravitational waves, reassuringly there is indirect (and very compelling) evidence for gravitational waves. A binary system of stars rotating around its common center of mass should radiate gravitational waves (much like electro-magnetic synchroton radiation). For two stars of equal mass M at distance $2r$ from each other, the prediction of General Relativity is that the power radiated by the binary system according to the general formula (23.118) is

$$P = dE/dt = -\frac{2}{5}\frac{G_N^4 M^5}{r^5} \quad . \tag{23.119}$$

This energy loss has actually been observed. In 1974, Hulse and Taylor discovered a binary system, affectionately known as PSR1913+16, in which both stars are small neutron stars, both roughly of solar mass, one of them being a *pulsar*, a rapidly spinning neutron star. The period of the orbit is only eight hours, and the fact that one of the stars is a pulsar provides a highly accurate clock with respect to which a change in the period as the binary loses energy can be measured. The observed value is in good agreement with the theoretical prediction for loss of energy by gravitational radiation and Hulse and Taylor were rewarded for these discoveries with the 1993 Nobel Prize. These observations have been confirmed and refined by the discovery and precise measurements and observations of other (even more extreme) binary systems.

Other situations in which gravitational waves might be either detected directly or inferred indirectly are extreme situations like gravitational collapse (supernovae) or matter orbiting black holes.

2016 Update[68]

On February 11, 2016, the LIGO Scientific Collaboration and Virgo Collaboration teams announced that they had made the first observation of gravitational waves, originating from a pair of merging black holes using the Advanced LIGO detectors. On June 15, 2016, a second detection of gravitational waves from coalescing black holes was announced.

[67] If you want (your PC) to contribute to the search for (continous) gravitational wave sources such as pulsars, take a look at the Einstein@Home project at `http://einstein.phys.uwm.edu/`.

[68] Quoted from `https://en.wikipedia.org/wiki/Gravitational_wave`. For more details, see e.g. `https://en.wikipedia.org/wiki/First_observation_of_gravitational_waves`.